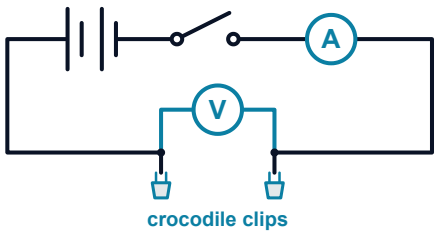
DO NOW, REQUIRED PRACTICAL

Resistance and the length of a wire

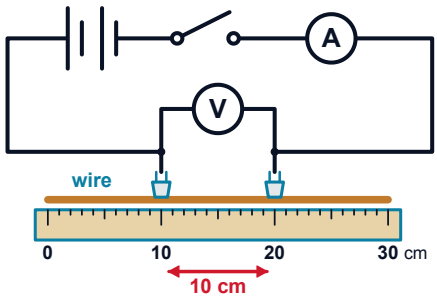
Electricity
Paper 1, Combined and Separate
6 steps

A student investigates how the length of a wire affects its resistance.

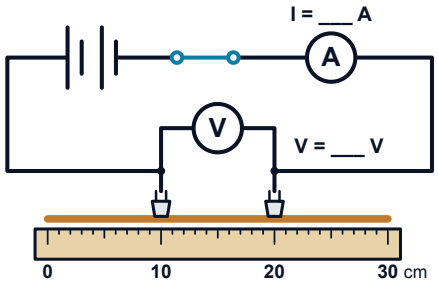
1 **Set up a circuit with an ammeter in series, ending in two crocodile clips. Connect a voltmeter across the clips.**



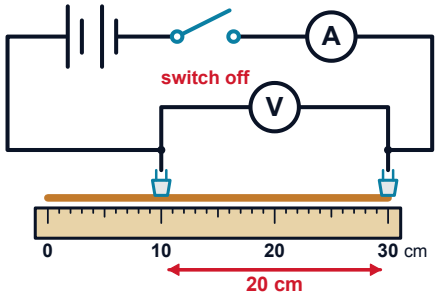
2 **Attach the crocodile clips 10 cm apart on the wire, so the voltmeter is across this length of wire.**



3 **Switch on and record the current and the potential difference.**



4 **Switch off to let the wire cool, then move the clips to 20 cm apart.**



5 **Repeat the readings for a range of lengths.**

length (cm)	I (A)	V (V)
5		
10		
15		
20		
25		
30		

6 **Calculate the resistance for each length and plot resistance against length.**

$$R = V \div I$$

(for each length)

